双电机扩展板

MG513XP28_12V ×2，配套CoralC5-Core
三串锂电池2800mAh，目标约一小时

核心板接口

L_PWM
L_DIR
MOTOR_RUN
L_nFAULT
L_ISENSE
L_ENC_A
L_ENC_B
V3V3
V5_SYS
R_PWM
R_DIR
R_nFAULT
R_ISENSE
R_ENC_A
R_ENC_B

文件：03_Core_Interface.kicad_sch

左电机与编码器

L_PWM
L_DIR
MOTOR_RUN
L_nFAULT
L_ISENSE
L_ENC_A
L_ENC_B
VM
V3V3

文件：04_Left_Motor.kicad_sch

右电机与编码器

R_PWM
R_DIR
MOTOR_RUN
R_nFAULT
R_ISENSE
R_ENC_A
R_ENC_B
VM
V3V3

文件：05_Right_Motor.kicad_sch

输入、总开关与降压

V5_SYS
VM

文件：02_Power.kicad_sch

拨动开关切断主电源
两路电机各自限流
编码器用三伏三

原理图设计：实板电流、温升与线束待验证

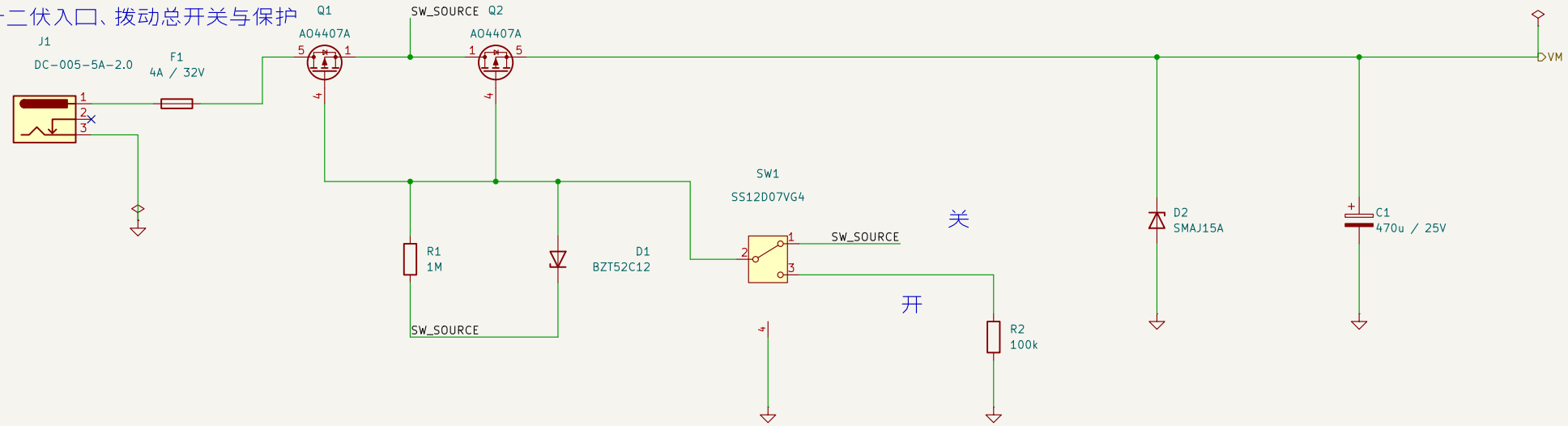
SmallCoral / CoralC5-Motor

Sheet: /
File: CoralC5-Motor.kicad_sch

Title: Coral-C5 双电机扩展板总览

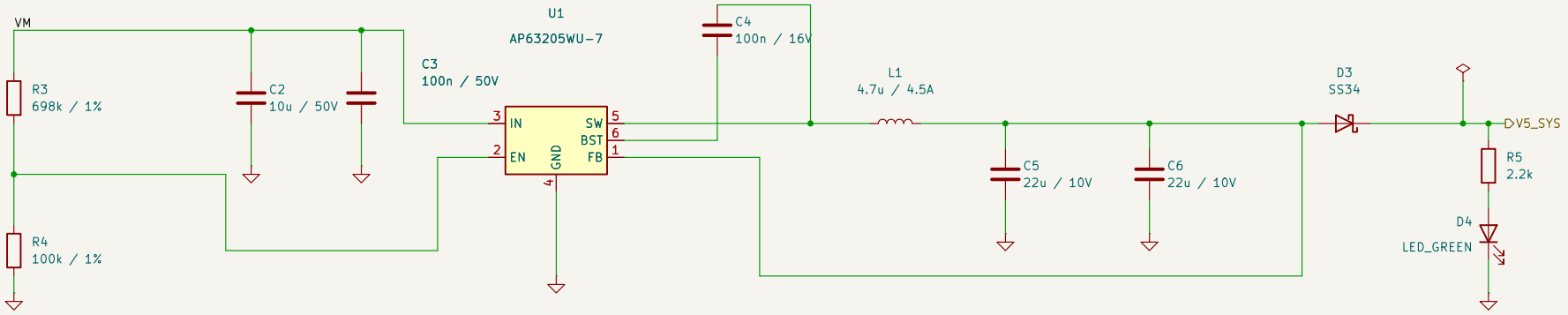
Size: A4	Date: 2026-11-03	Rev: M1.1
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十二伏入口、拨动总开关与保护



三串锂电池9~12.6V，中心正极；拨动开关控制双MOS主通路
关闭后电机和降压均断电，USB仍可单独供核心板

五伏同步降压与USB供电隔离



五伏负载预算0.75A，核心板入口约4.6~4.9V
按原厂配置4.7uH及两颗22uF，D3隔离USB反向供电

原理图设计：实板电流、温升与线束待验证

SmallCoral / CoralC5-Motor

Sheet: /输入、总开关与降压/
File: 02_Power.kicad_sch

Title: 十二伏入口、拨动总开关与五伏降压

Size: A4	Date: 2026-11-03	**Rev: M1.1**
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核心板接口

J14A

C5-EDGE-3 / 98P

GND	A1	A1_/_GND	B1	B1_/_GND	GND
AUX_UART_TX	A2	A2_/_PA2	B2	B2_/_PA3	AUX_UART_RX
L_ENC_B	A3	A3_/_PA1	B3	B3_/_PA4	AUX_ADC
L_ENC_A	A4	A4_/_PA0	B4	B4_/_PA5	AUX_GPIO
L_ISENSE	A5	A5_/_PH5	B5	B5_/_PA6	
	A6	A6_/_VREF_MCU	B6	B6_/_PA7	
L_nFAULT	A7	A7_/_PH4	B7	B7_/_PC4	
L_DIR	A8	A8_/_PC3	B8	B8_/_PC5	
L_PWM	A9	A9_/_PC2	B9	B9_/_PB0	
	A10	A10_/_PC1	B10	B10_/_PB1	
R_ISENSE	A11	A11_/_PC0	B11	B11_/_PB2	
NRST	A12	A12_/_NRST	B12	B12_/_PE7	
	A13	A13_/_NC	B13	B13_/_PE8	MOTOR_EN_REQ
	A14	A14_/_NC	B14	B14_/_PE9	
	A15	A15_/_PC15	B15	B15_/_PE10	
	A16	A16_/_PC14	B16	B16_/_PE11	
	A17	A17_/_PC13	B17	B17_/_PE12	
	A18	A18_/_PH15	B18	B18_/_PE13	
	A19	A19_/_PE6	B19	B19_/_PE14	
	A20	A20_/_PE5	B20	B20_/_PE15	
	A21	A21_/_PE4	B21	B21_/_PB10	
	A22	A22_/_PE3	B22	B22_/_PB12	
	A23	A23_/_PE2	B23	B23_/_PB13	
	A24	A24_/_PE1	B24	B24_/_PB14	
	A25	A25_/_PE0	B25	B25_/_PB15	

J14B

C5-EDGE-3 / 98P

	A26	A26_/_PB9	B26	B26_/_PD8	
R_PWM	A27	A27_/_PB8	B27	B27_/_PD9	
	A28	A28_/_PH2	B28	B28_/_PD10	
AUX_I2C_SDA	A29	A29_/_PB7	B29	B29_/_PD11	
AUX_I2C_SCL	A30	A30_/_PB6	B30	B30_/_PD12	
	A31	A31_/_PB5	B31	B31_/_PD13	
	A32	A32_/_PB4	B32	B32_/_PD14	
R_ENC_A	A33	A33_/_PB3	B33	B33_/_PD15	
R_DIR	A34	A34_/_PD7	B34	B34_/_PC6	
R_nFAULT	A35	A35_/_PD6	B35	B35_/_PC7	
	A36	A36_/_PD5	B36	B36_/_PC8	
	A37	A37_/_PD4	B37	B37_/_PC9	
	A38	A38_/_PD3	B38	B38_/_PA8	
	A39	A39_/_PD2	B39	B39_/_PA9	
	A40	A40_/_PD1	B40	B40_/_PA10	
	A41	A41_/_PD0	B41	B41_/_PA11	
	A42	A42_/_PC12	B42	B42_/_PA12	
	A43	A43_/_PC11	B43	B43_/_PA13	SWDIO
	A44	A44_/_PC10	B44	B44_/_PH3	
R_ENC_B	A45	A45_/_PA15	B45	B45_/_NC	
SWCLK	A46	A46_/_PA14	B46	B46_/_NC	
V3V3	A47	A47_/_+3V3	B47	B47_/_+3V3	V3V3
V5_SYS	A48	A48_/_+5V_SYS	B48	B48_/_+5V_SYS	V5_SYS
GND	A49	A49_/_GND	B49	B49_/_GND	GND

调试与扩展

J14C

C5-EDGE-3 / 98P

GND	50	MP1_/_GND MP2_/_GND	51	GND

J8

SWD / XH2.50

V3V3	1
SWDIO	2
SWCLK	3
GND	4

硬件停机回路

J10

RUN / STOP_NC

R6

100R

MOTOR_EN_REQ

MOTOR_RUN

R7

10k

MOTOR_RUN

J10装短路帽，或接常闭停机开关
开路、复位时默认停止电机

V3V3

严格沿用C5-EDGE-3针序
空闲触点明确标记不连接
该母座不能插计算机PCIe

编码器及三伏三外接负载
合计不得超过100mA

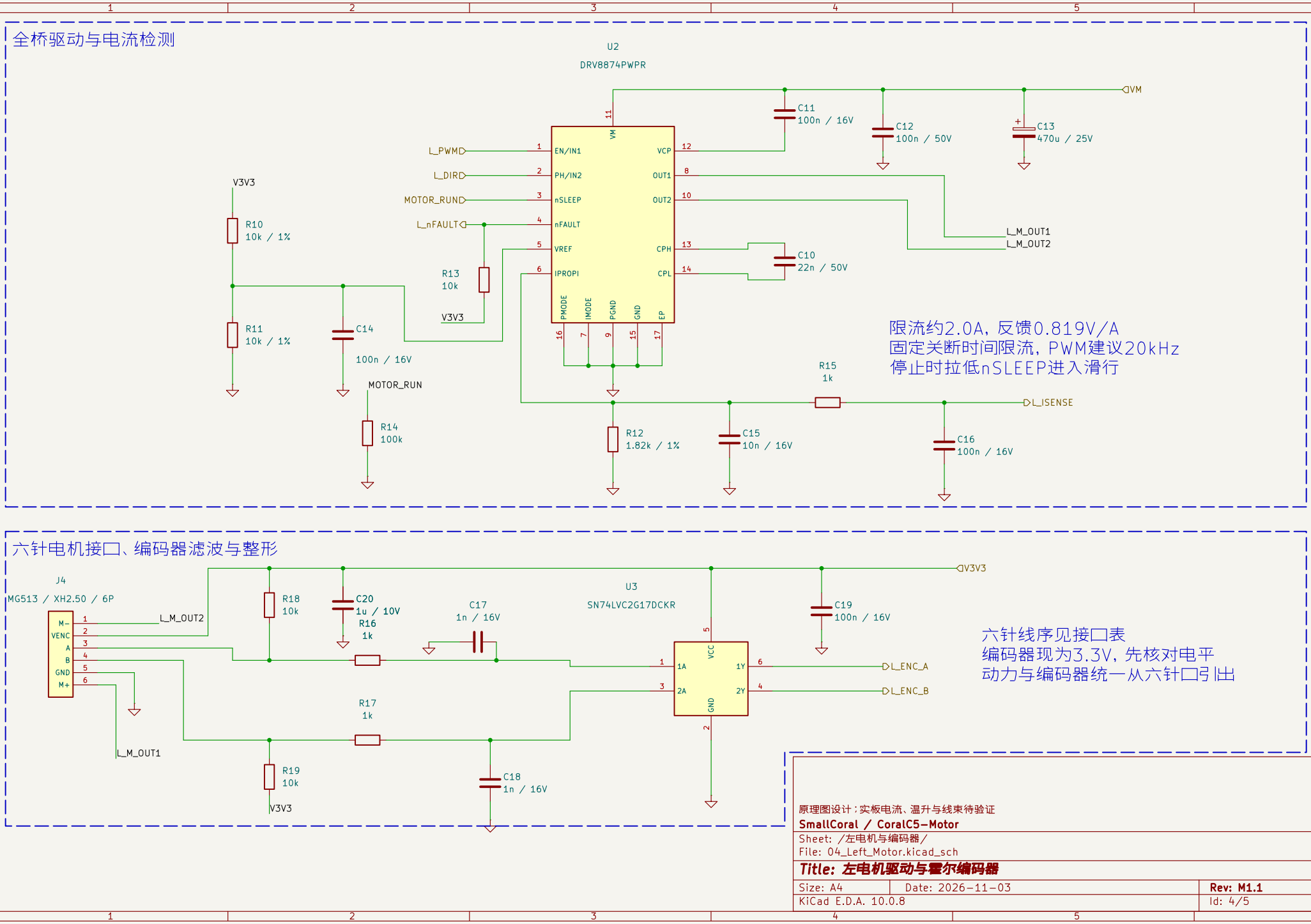
原理图设计：实板电流、温升与线束待验证

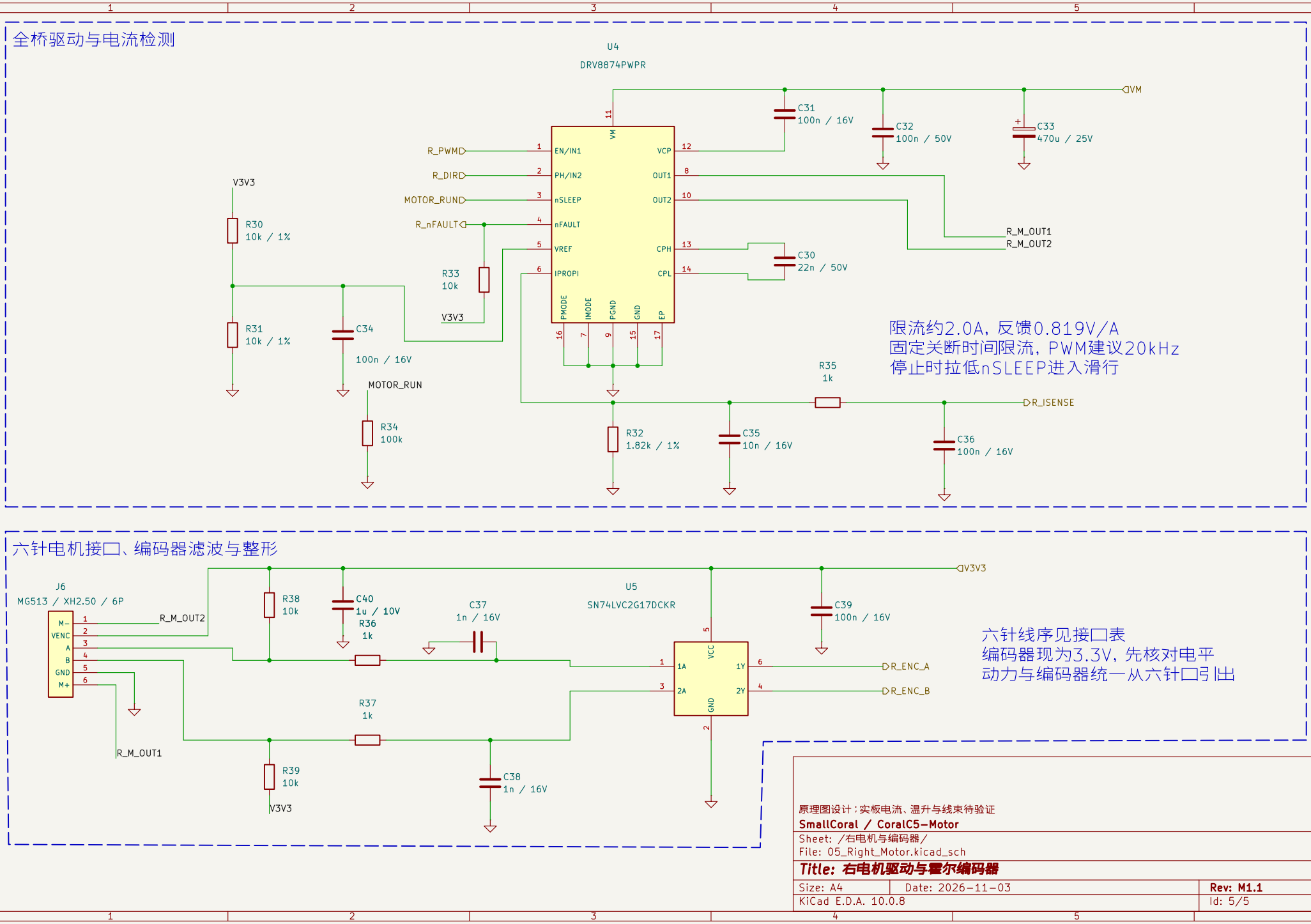
SmallCoral / CoralC5-Motor

Sheet: /核心板接口/
File: 03_Core_Interface.kicad_sch

Title: C5-EDGE-3 核心板接口与调试

Size: A4	Date: 2026-11-03	**Rev: M1.1**
KiCad E.D.A. 10.0.8		Id: 3/5


全桥驱动与电流检测
U2
DRV8874PWPR
VM
C11
100n / 16V
C12
100n / 50V
C13
470u / 25V
L_PWM
L_DIR
MOTOR_RUN
L_nFAULT
EN/IN1
PH/IN2
nSLEEP
nFAULT
VREF
IPROPI
VCP
OUT1
OUT2
CPH
CPL
PMODE
IMODE
PGND
GND
EP
L_M_OUT1
L_M_OUT2
C10
22n / 50V
V3V3
R10
10k / 1%
R11
10k / 1%
R13
10k
C14
100n / 16V
MOTOR_RUN
R14
100k
限流约2.0A，反馈0.819V/A
固定关断时间限流，PWM建议20kHz
停止时拉低nSLEEP进入滑行
R15
1k
L_ISENSE
R12
1.82k / 1%
C15
10n / 16V
C16
100n / 16V
六针电机接口、编码器滤波与整形
J4
MG513 / XH2.50 / 6P
M-
VENC
A
B
GND
M+
L_M_OUT2
L_M_OUT1
R18
10k
C20
1u / 10V
R16
1k
C17
1n / 16V
U3
SN74LVC2G17DCKR
VCC
1A
2A
1Y
2Y
GND
C19
100n / 16V
L_ENC_A
L_ENC_B
六针线序见接口表
编码器现为3.3V，先核对电平
动力与编码器统一从六针口引出
R17
1k
R19
10k
C18
1n / 16V
原理图设计：实板电流、温升与线束待验证
SmallCoral / CoralC5-Motor
Sheet: /左电机与编码器/
File: 04_Left_Motor.kicad_sch
Title: 左电机驱动与霍尔编码器
Size: A4
Date: 2026-11-03
Rev: M1.1
KiCad E.D.A. 10.0.8
Id: 4/5


全桥驱动与电流检测
U4
DRV8874PWPR
VM
C31
100n / 16V
C32
100n / 50V
C33
470u / 25V
R_PWM
R_DIR
MOTOR_RUN
R_nFAULT
EN/IN1
PH/IN2
nSLEEP
nFAULT
VREF
IPROPI
VCP
OUT1
OUT2
CPH
CPL
PMODE
IMODE
PGND
GND
EP
R_M_OUT1
R_M_OUT2
C30
22n / 50V
V3V3
R30
10k / 1%
R31
10k / 1%
C34
100n / 16V
R33
10k
MOTOR_RUN
R34
100k
限流约2.0A，反馈0.819V/A
固定关断时间限流，PWM建议20kHz
停止时拉低nSLEEP进入滑行
R35
1k
R_ISENSE
R32
1.82k / 1%
C35
10n / 16V
C36
100n / 16V
六针电机接口、编码器滤波与整形
J6
MG513 / XH2.50 / 6P
M-
VENC
A
B
GND
M+
R_M_OUT2
R_M_OUT1
R38
10k
C40
1u / 10V
R36
1k
C37
1n / 16V
U5
SN74LVC2G17DCKR
VCC
1A
2A
1Y
2Y
GND
C39
100n / 16V
R_ENC_A
R_ENC_B
六针线序见接口表
编码器现为3.3V，先核对电平
动力与编码器统一从六针口引出
R37
1k
R39
10k
C38
1n / 16V
原理图设计：实板电流、温升与线束待验证
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Id: 5/5